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Novel insight into enhanced recoverability of acidic inhibition to anaerobic digestion with nano-bubble water supplementation

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HIGHLIGHTS

- The acidic inhibition was alleviated under nano-babble water (NBW) addition.
- Digestion stability under acidic inhibition was improved by adding NBW.
- CH_4 production was enhanced by 12–17% by NBW addition even under acid inhibition.
- NBW could promote the production/consumption of VFAs even under acidic inhibition.
- NBW addition increased the maximum CH_4 production rate.

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ABSTRACT

Nano-bubble water (NBW) has been proven to be effective in promoting organics utilization and CH_4 production during anaerobic digestion (AD) process, suggesting its potential in improving the stability of the AD process and thereby alleviating acidic inhibition. In this work, the effect of NBW on digestion stability and CH_4 production was investigated to evaluate the ability of NBW on AD recovery from acidic inhibition. Results showed that NBW supplementation increased the total alkalinity (TA) and partial alkalinity (PA), and reduced the ratio of VFA/TA, thus maintained the stability of the AD process. Generation/consumption of VFAs was also enhanced with NBW supplementation under acidic inhibition with pH values of 5.5, 6.0 and 6.5. The cumulative CH_4 production was 246–257 mL/g-VS in NBW groups, which was 12.1–17.2% higher than the control. Moreover, with NBW supplementation, the maximum CH_4 production rate was raised according to the modeling results.

1. Introduction

Anaerobic digestion (AD), a biological process, has been used to treat solid organic wastes including animal manure due to various advantages including biogas and CH_4 generation (Zhang et al., 2019). In addition, elimination of harmful pathogens provides the added benefit in terms of low contamination to the surrounding environment via AD (Ueki et al., 2018). Therefore, the treatment of organic feedstock by AD is promising to produce clean and renewable energy as well as protecting the environment.

Performance of AD system could be influenced by many factors like intermediate products, inoculation ratio, total solids (TS) content, temperature and initial pH (Cao et al., 2020). pH drop is one of the component factors that trigger hindrance of methanogenesis during AD. The shift in pH would lead to variation in the dominant microbial community and affect the main organic acids products (Zhang et al., 2015). Low pH during AD is one of the main factors that cause methanogens to inhibit. This is because the minimum pH value for methanogens to maintain high activity needs to be not <6.5 (Wang et al., 2014). Rapid pH drops in the early stages of AD, ascribed to the accumulation of volatile fatty acids (VFAs) or long-chain fatty acid, would result in reduction of biological activity and finally restrict AD process (Li et al., 2017; Liu et al., 2015). As reported by previous studies, optimizing the initial pH with additional chemicals (sodium hydroxide or hydrochloric acid) would be useful to maintain the stability of AD process (Zhai et al., 2015; Lin et al., 2013). Although the addition of

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external chemicals is beneficial to the AD process, the addition not only increases costs but also environmentally unsound. The additional facility which is protected for the surrounding environment against harmful chemicals is required. New technologies are desired to overcome the obstruction of pH drop to AD, maintain digestion stability and CH_4 production meanwhile under abnormal pH conditions (acidic/alkaline inhibition conditions) without additional chemicals.

Nano-bubble water (NBW) which contains enormous nanoscale particle bubbles, possesses many specific properties like higher mobility and gas solubility superior to ordinary water or deionized water (DW) (Liu et al., 2019). Besides, reactive oxygen species (ROSs), including singlet oxygen ($^1O_2$), superoxide anion radicals ($O_2^{\bullet-}$), and hydroxyl radicals (•OH) would be generated during the collapsing of nano-bubbles (NBs) (Lyu et al., 2019). Moreover, the numerous ROSs produced during NBs collapse may promote the hydrolysis of organics, especially the hydroxyl radicals with relatively higher redox potential and have been confirmed to promote the degradability of various recalcitrant organics (Khan et al., 2014). In addition, NBW supplementation would also improve the mass transferability (Temesgen et al., 2017). Due to these unique characteristics, NBW has been successfully applied in many fields, such as environmental toxicology (Fan et al., 2020a), functional food (Guo et al., 2020b), construction industry (Kim et al., 2020), biofilm (Xiao and Xu, 2020) and agriculture (Zhou et al., 2020). Most recently, NBW has been demonstrated to be effective in promoting CH_4 production during AD process (Yang et al., 2019, Fan et al., 2020c), which could be ascribed to the enhanced activity of hydrolytic and methanogenic bacteria (Wang et al., 2020). Previous studies have shown that different kinds of NBW (Air, N_2, CO_2, or H_2) can all increase CH_4 production in the AD process (Wang et al., 2019, Fan et al., 2020b). However, it's noteworthy that the above studies were performed with no inhibition happened during AD, whether NBW is beneficial to improve CH_4 production and AD stability under acidic inhibition is still unknown. Considering the potential of NBW in promoting hydrolysis of organics during AD, it would be of great promising in facilitating the recovery of AD performance under acidic inhibition. Moreover, as an environmentally friendly and cost-effective method (easy availability of NBW and no chemical addition), the supplementation of NBW for acidic inhibition alleviation would be of great prospect in practical application.

Therefore, the purpose of this work will be threefold: firstly, to study the effect of NBW supplementation on the stability of AD performance under acidic inhibition conditions, secondly, to evaluate the CH_4 production of NBW supplementation under acidic inhibition conditions, and thirdly, to analyze the possible mechanisms and application potential involved. This study is the first work to investigate the effects of NBW supplementation on digestion stability, CH_4 production during AD process under acidic inhibition. Results of this study is targeting a new application of NBW in AD process under acidic inhibition conditions.

2. Materials and methods

2.1. Preparation and characterization of NBW

NBW used in this research was prepared by using a nano-bubble generator (HACK FB11, Japan) based on the previous reports (Wang et al., 2019). Air in the natural environment was introduced into DW through this generator at 20 ± 2 °C. NBW was produced by a combination of rotational flow, pressurized mechanical cyclic, and cavitation. Shearing was performed while pressurizing to generate cavitation, and then bubbles were generated through a bubbling nozzle (Wang et al., 2019). After this nano-bubble generator was operated for 20 min, the air-NBW was obtained and applied to the AD process in this work. The characteristics of air-NBW and DW were tested for the purpose of better comparison. The zeta potential of DW and air-NBW was measured by using a zeta potential analyzer (Malvern, Nano ZS, United Kingdom). The pH of air-NBW and DW was tested by a pH meter (METTLER TOLEDO FE20, Switzerland). The nano-bubble size and concentration were detected using the nanoparticle tracking analysis method (Malvern, NanoSight LM10, UK).

Table 1
Characteristics of raw swine manure and inoculum used in this study.*

Parameters	Method	Unit	Inoculum	Raw pig manure
Total solids	APHA, 2012	%	1.19 ± 0.04	32.03 ± 0.75
Volatile solids	APHA, 2012	%	0.96 ± 0.03	27.25 ± 0.09
Total ammonia nitrogen	APHA, 2012	mg/g	0.39 ± 0.03	3.44 ± 0.12
Total alkalinity (as CaCO3)	Sun et al., 2016	mg/g	2.38 ± 0.04	18.08 ± 0.56
Soluble total carbon	TOC analyzer	mg/g	1.06 ± 0.02	40.47 ± 0.89
Soluble inorganic carbon	TOC analyzer	mg/g	0.50 ± 0.02	3.71 ± 0.27
Soluble total organic carbon	TOC analyzer	mg/g	0.57 ± 0.01	36.77 ± 1.17
Volatile fatty acids	Gas chromatograph	mg/g	0.13 ± 0.01	36.30 ± 0.84
Soluble proteins	Lowry method	mg/g	ND**	11.23 ± 0.64
Soluble carbohydrates	Phenol-sulfuric	mg/g	ND**	6.03 ± 0.24
C	Elemental analyzer	%/TS	40.00	44.73
H	Elemental analyzer	%/TS	6.24	6.82
N	Elemental analyzer	%/TS	3.21	4.04
S	Elemental analyzer	%/TS	0.85	0.67
C/N ratio		–	12.46	11.07
pH	pH meter	–	7.08 ± 0.04	6.17 ± 0.05

* Data in the table are expressed as Means ± SD.
** ND (Not Determined).

2.2. Batch anaerobic fermentation experiments

2.2.1. Raw materials and inoculum

In this work, raw pig manure (PM) sampled from a domestic pig farm in Tsukuba, Ibaraki, Japan was used as the substrate. After manually sorting out impurities such as plastics and small stones, the raw PM was placed in a 4 °C refrigerator. Meantime, this raw PM was thoroughly mixed to ensure uniformity before subsequent AD experiments. Anaerobically digested sludge was obtained from a wastewater treatment plant in Ibaraki, Japan. The digested sludge was cultivated using 1 L Scott Duran bottles with the raw PM (collected from the same pig farm) about one month until no biogas production for acclimation of the inoculum. Characteristics of PM and inoculum are presented in Table 1 according to five tests in parallel.

2.2.2. Batch AD experiments procedures

The suitable pH level for methanogens ranges from 6.8 to 7.2 (Zhou et al., 2016). When the pH level is lower than 5.3, it is not conducive to CH_4 production (Wu et al., 2010). Therefore, this work simulated the AD process under acidic inhibition with pH values of 5.5, 6.0 and 6.5, respectively. All batch AD experiments in this work were conducted in 250 mL Scott Duran bottles. To simulate anaerobic fermentation under initial acidic inhibition conditions, 2.0 M HCl or NaOH was added to adjust the initial pH value at the beginning of the AD experiments. Six groups of reactors (R) were designed: pH in $R(6.5\ pH)_{NBW}$/$R(6.5\ pH)_{Control}$, $R(6.0\ pH)_{NBW}$/$R(6.0\ pH)_{Control}$ and $R(5.5\ pH)_{NBW}$/$R(5.5\ pH)_{Control}$ were adjusted to 6.5, 6.0 and 5.5, respectively. 10 g PM and 90 g inoculum were loaded in each reactor, additionally, 100 g DW was

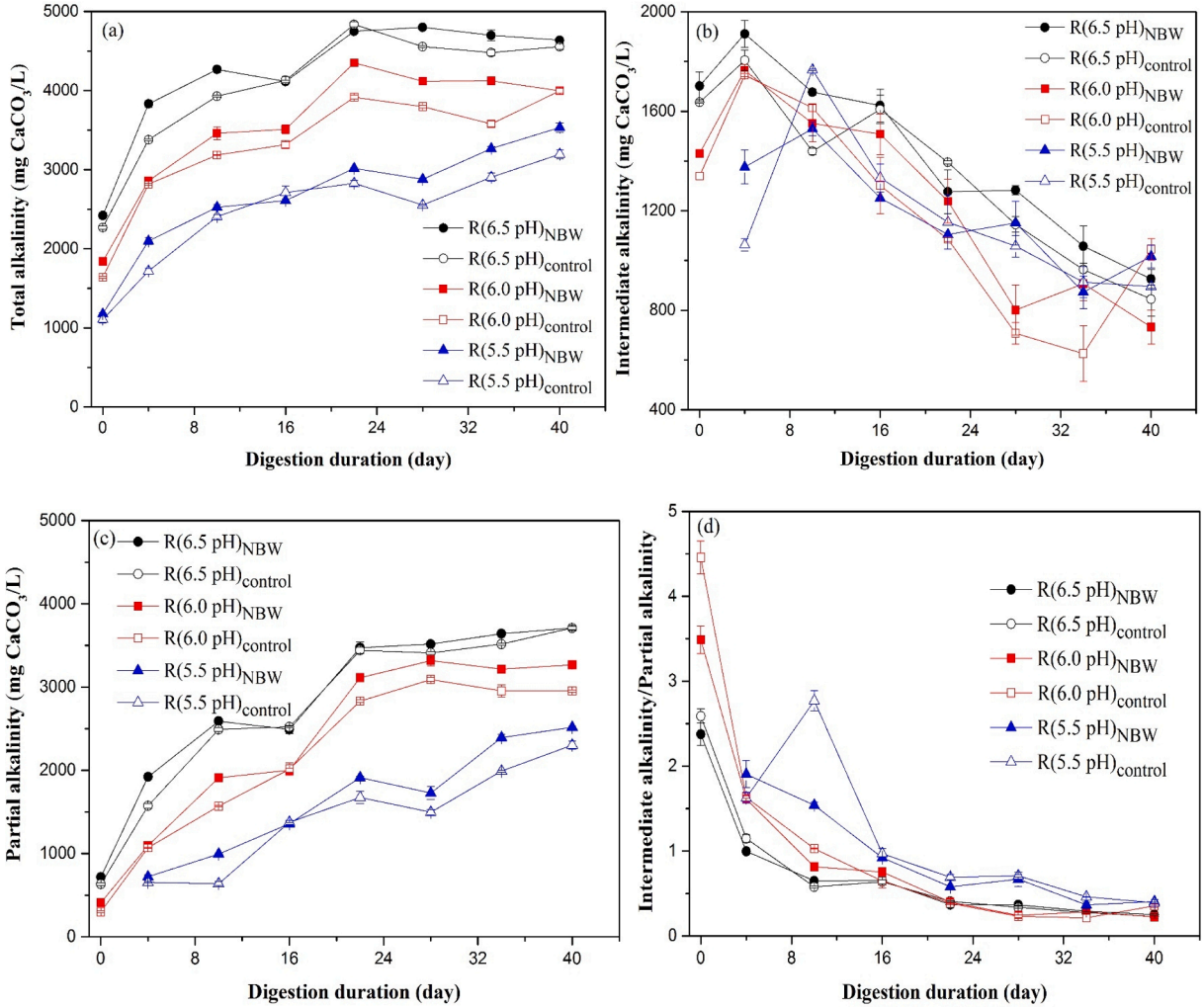


Fig. 1. (a) Total alkalinity variation during AD process. (b) Intermediate alkalinity variation during AD process. (c) Partial alkalinity variation during AD process. (d) Intermediate alkalinity/Partial alkalinity variation during AD process.

added into $R_{Control}$ groups while the same weight of NBW was added into R_{NBW} groups at the beginning of the AD experiments. Rubber stoppers and aluminum caps were used to seal all reactors after all materials were loaded. All reactors were flushed with nitrogen for 3 min and then placed in a temperature-controlled incubator to maintain the temperature at 35 ± 2 °C. The digestion duration was 40 days according to the preliminary experiments for all reactors and all bottles would be shaken for 2 min at a fixed time every day. All experiments were carried out triplicate.

2.2.3. Analytical methods

The biogas produced from each reactor was determined every two days using a gas-tight 60 mL syringe. Gas compositions (CH_4 + CO_2 + H_2) were measured by a gas chromatograph (GC-8A/TCD, SHIMADZU CO., Japan). Volatile solids (VS), TS and total ammonia nitrogen (TAN) were measured according to the standard method (APHA, 2012). A pH meter (METTLER TOLEDO FE20, Switzerland) was used for determination of pH value. Before grinding into powder, the raw PM and inoculum were dried at 105 °C to analyze C, N, H, and S by using an elemental analyzer (PerkinElmer 2400 II CHN Elemental Analyzer, USA). The concentration of soluble proteins was measured by using the Lowry method with bovine serum albumin (BSA) as the standard. Total alkalinity (TA), intermediate alkalinity (IA) and partial alkalinity (PA) were determined based on a previous report (Sun et al., 2016). Alkalinity measurements to pH endpoints of 5.75 and 4.3 were applied to evaluate AD stability, among which, PA (to pH 5.75) and IA (from pH 5.75 to 4.3). Soluble total carbon (STC), soluble inorganic carbon (SIC), and soluble total organic carbon (STOC) in the filtrate were measured by a TOC analyzer (TOC-VCSN, ASI-V autosampler, SHIMADZU CO., Japan). The concentration of soluble carbohydrates was measured by phenol–sulfuric method with glucose (+) as the standard. The gas chromatograph (GC-8A, SHIMADZU CO., Japan) with flame ionization detector (FID) and Unisole F-200 30/60 column was used to analyze volatile fatty acids (VFAs).

2.3. Calculation methods

VS reduction rate (%) was calculated based on the Eq. (1)

$$\text{VS reduction rate}(\%) = \frac{[\text{VS}]_{\text{initial}} - [\text{VS}]_{\text{final}}}{[\text{VS}]_{\text{initial}}} \times 100 \quad (1)$$

2.4. Kinetic models

In this work, the modified Gompertz model (Eq. (2)) was used to analyze the kinetics of the AD process under three different acidic inhibition conditions with or without NBW supplementation.

$$P = P_{max} exp\left\{ - exp\left[\frac{R_{max} e}{P_{max}}(\lambda - t) + 1\right]\right\} \quad (2)$$

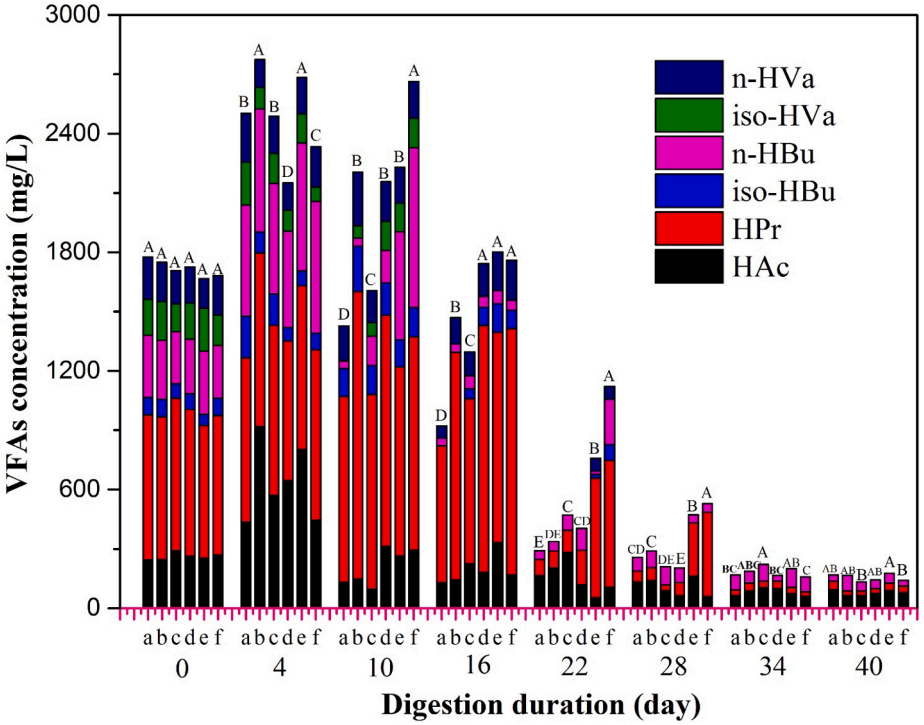


Fig. 2. VFAs concentration variation during the whole AD process. The alphabets under X-axis areas: a-R(6.5 pH)$_{NBW}$, b-R(6.5 pH)$_{Control}$, c-R(6.0 pH)$_{NBW}$, d-R(6.0 pH)$_{Control}$, e-R(5.5 pH)$_{NBW}$, f-R(5.5 pH)$_{Control}$. One-way analysis of variance (ANOVA) followed by Tukey's post-hoc test was applied for the statistical analysis of total VFAs. The different letters (A, B, C, …) indicate significant difference at the 5% level ($p < 0.05$).

where P is the cumulative CH_4 production (mL/g-VS), R_{max} is the maximum CH_4 production rate (mL/g-VS·d), P_{max} is the maximum CH_4 production potential (mL/g-VS), λ is the lag phase (d).

2.5. Statistical analysis

The mean value ± standard deviation based on three parallel experiments were calculated by using Microsoft Office Excel 2019 for AD experiments and NBW characteristics tests. One-way analysis of variance (ANOVA) followed by Tukey's post-hoc test was performed to evaluate the statistical differences regarding performance indicators. The different letters (A, B, C, …) indicate significant difference at the 5% level ($p < 0.05$). In this work, Microsoft Office Excel 2019 and Origin 9.0 software were used for statistical analysis, and for the analysis of the modified Gompertz model.

3. Results and discussion

3.1. Performance of AD process

Alkalinity, including TA, IA and PA can be used as stability indicators for AD process (Sun et al., 2019). Among which, PA represents the buffering from ammonia and bicarbonate while IA shows the influence from VFA (Sun et al., 2016). The changes in TA of all reactors are shown in Fig. 1a. Overall, value of TA content varied in different groups and followed the order: R(6.5 pH) > R(6.0 pH) > R(5.5 pH). The declined TA content in reactors with lower initial pH value could be ascribed to the larger amount of acidic substances (2.0 M HCl) addition for pH adjustment. As for the progressing of fermentation, TA content in all reactors increased gradually, indicating the recovery of digestion stability. In addition, in all the reactors of R(6.5 pH), R(6.0 pH) or R(5.5 pH), significantly higher TA content was observed in R_{NBW} than the corresponding $R_{Control}$ on the 4th and 10th day ($p < 0.05$), indicating that under acidic inhibition, the supplementation of NBW could increase the total alkalinity (TA) content in the initial stage of AD, which may play an important role in the recovery of the stability of the acidic AD system.

Fig. 1b and Fig. 1c show the content of IA and PA, respectively. The IA content experienced a short-term increase and then gradually decreased. Since IA is affected by VFAs, the change of IA content is related to the content of VFAs. The content of VFAs also experienced a short-term increase and then a rapid decrease (Fig. 2), which was consistent with the change of IA content. On the contrary, the PA content gradually increased with the AD process progressed, which indicated that the buffering capacity of ammonia and bicarbonate in all reactors increased. It was worth noting that on the 10th day, the PA content in R_{NBW} was significantly higher than the corresponding $R_{Control}$ ($p < 0.05$).

When IA/PA increases suddenly, the buffering capacity of the AD system is weakened (Sun et al., 2016). The IA/PA ratio under different acidic inhibition conditions is shown in Fig. 1d. The initial IA/PA ratio of all reactors was greater than 1.5, indicating that the stability of all reactors was low in the initial state. As the AD process progressed, the IA/PA ratio of all reactors was rapidly decreasing, except for R(5.5 pH)$_{Control}$. As for R(5.5 pH)$_{Control}$, it experienced a trend of increasing first and then decreasing, which indicated that the stability of R(5.5 pH)$_{Control}$ first deteriorated and then gradually recovered, while the other reactors gradually recovered their stability (Fig. 1d). In addition, the IA/PA ratio of R_{NBW} decrease faster than the corresponding $R_{Control}$, especially in R(5.5 pH)$_{NBW}$. The IA/PA ratio of R(5.5 pH)$_{NBW}$ was significantly lower than the corresponding R(5.5 pH)$_{Control}$ on the 10th day ($p < 0.05$). These results indicated that the supplementation of NBW was beneficial to the restoration of the acidic AD system.

VFAs as key intermediate metabolites in AD process are products by hydrolysis and acidogenesis. The stability of VFAs represents the balance between methanogenesis and hydrolysis/acidogenesis (Li et al., 2015). The VFAs concentrations in all reactors are shown in Fig. 2. The VFAs concentrations increased first and then decreased and were lower than 600 mg/L after 28 days. On the 4th day, the VFAs concentrations of R(6.0 pH)$_{NBW}$ and R(5.5 pH)$_{NBW}$ were significantly higher than that of R

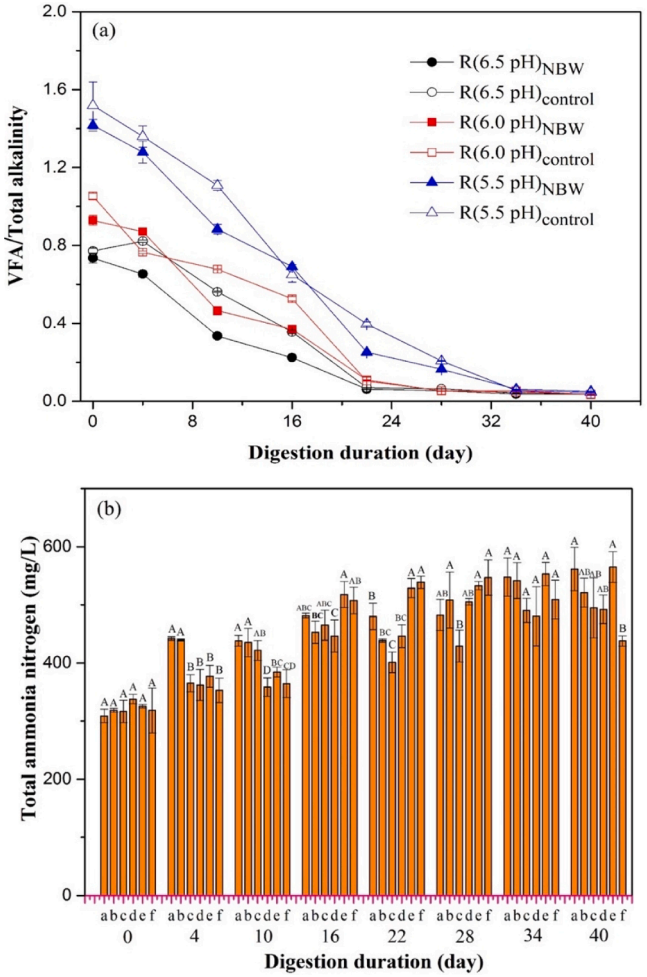


Fig. 3. (a) VFA/Total alkalinity. (b) Total ammonia nitrogen. The alphabets under X-axis areas: a-R(6.5 pH)$_{NBW}$, b-R(6.5 pH)$_{Control}$, c-R(6.0 pH)$_{NBW}$, d-R(6.0 pH)$_{Control}$, e-R(5.5 pH)$_{NBW}$, f-R(5.5 pH)$_{Control}$. One-way analysis of variance (ANOVA) followed by Tukey's post-hoc test was applied for the statistical analysis of total ammonia nitrogen. The different letters (A, B, C, …) indicate significant difference at the 5% level ($p < 0.05$).

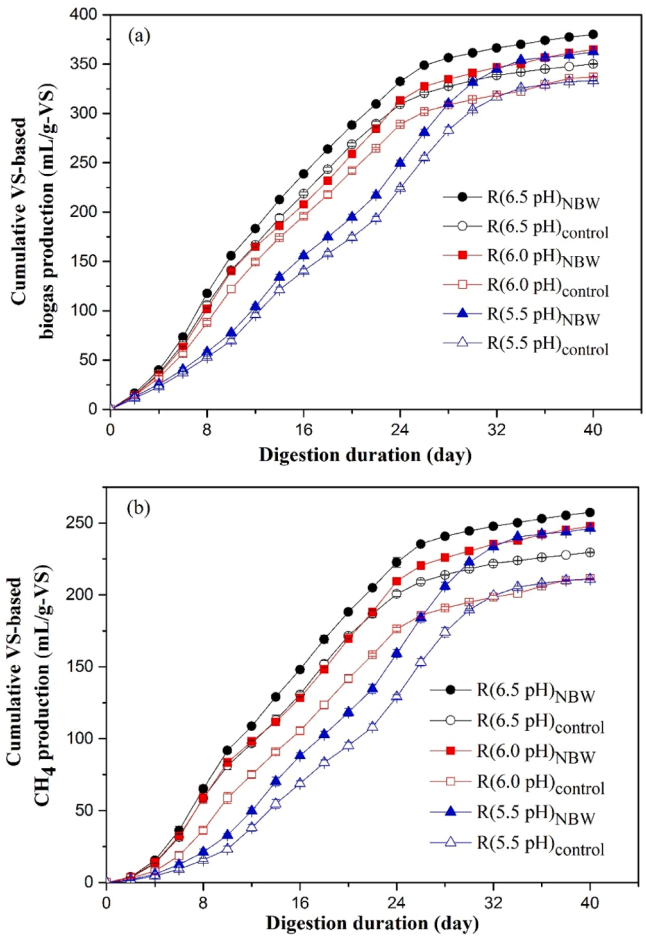


Fig. 4. (a) Cumulative VS-based biogas production. (b) Cumulative VS-based CH_4 production.

(6.0 pH)$_{Control}$ and R(5.5 pH)$_{Control}$, respectively ($p < 0.05$). This phenomenon indicated that the supplementation of NBW promote the production of VFAs in the hydrolysis-acidification stage. From day 4 to day 16, the VFAs concentrations of R(6.5 pH)$_{NBW}$ were significantly lower than that of R(6.5 pH)$_{Control}$ ($p < 0.05$), while the biogas and CH_4 production of R(6.5 pH)$_{NBW}$ were higher than that of R(6.5 pH)$_{Control}$ (Fig. 4). Moreover, compared with R(6.5 pH)$_{Control}$, R(6.5 pH)$_{NBW}$ has significantly higher TA content on the 4th and 10th day ($p < 0.05$), which indicated that the lower VFAs concentration of R(6.5 pH)$_{NBW}$ was not due to less production but more consumption. Through this work, it could be concluded that even under acidic inhibition conditions, the supplementation of NBW promote the production and consumption of VFAs in the hydrolysis-acidification stage, which was beneficial to the subsequent methanation process.

As an important indicator to evaluate the stability of the AD process, the VFA/TA ratio has been applied in the AD system (Chen et al., 2014), and a ratio higher than 0.4 indicates the unstable AD process (Adou et al., 2020). Fig. 3a shows the VFA/TA ratio with or without NBW supplementation under three different acidic inhibition conditions. In the initial stage, the VFA/TA ratio of all reactors was greater than 0.7. Since R(5.5 pH) was greater than 1.4, the stability of R(5.5 pH) was relatively the worst in the initial stage. As the AD process progressed, the VFA/TA ratio of all reactors decreased, which was consistent with the IA/PA ratio (Fig. 1d). It took the longest time for R(5.5 pH)$_{Control}$ to be lower than 0.4, which indicated that the AD system stability of R(5.5 pH)$_{Control}$ was the worst relative to the other reactors. Moreover, the VFA/TA ratio of R_{NBW} was significantly lower than the corresponding $R_{Control}$ on the 10th day ($p < 0.05$), suggesting that supplementation of NBW enhanced the recovery of digestion stability.

The changes of TAN are shown in Fig. 3b. The TAN content of all reactors gradually increased during the initial stage of the AD process. This may be due to the decomposition of soluble organic nitrogen such as protein in the initial stage of AD, which increased the content of inorganic nitrogen including ammonia nitrogen. Bi et al. (2020) reported that when the TAN concentration exceeds 3,000 mg/L, the activity of methanogens will be affected and the AD process will be inhibited, resulting in a decrease in CH_4 production. In this work, the TAN content in all reactors was <1,000 mg/L (Fig. 3b). Therefore, no ammonia inhibition appeared in this study.

3.2. *CH_4 production performance*

The cumulative biogas production and cumulative CH_4 production of all reactors with NBW or DW supplementation under three acidic inhibition conditions are shown in Fig. 4. Overall, the cumulative CH_4 and biogas yields of R_{NBW} were higher than those of $R_{Control}$. As for the cumulative biogas production (Fig. 4a), R(6.5 pH)$_{NBW}$, R(6.0 pH)$_{NBW}$ and R(5.5 pH)$_{NBW}$ achieved 380 ± 5, 365 ± 6 and 363 ± 8 mL/g-VS, respectively, increasing by 8.5%, 8.2%, and 8.8% in comparison to R(6.5 pH)$_{Control}$, R(6.0 pH)$_{Control}$ and R(5.5 pH)$_{Control}$, respectively. Moreover, the cumulative biogas production of R_{NBW} was significantly higher than

Table 2
Kinetic parameters of cumulative CH_4 production from AD of SM with and without adding NBW under different acidic conditions.*

Model parameters	R(6.5 pH)$_{NBW}$	R(6.5 pH)$_{Control}$	R(6.0 pH)$_{NBW}$	R(6.0 pH)$_{Control}$	R(5.5 pH)$_{NBW}$	R(5.5 pH)$_{Control}$
R^2	0.9973	0.9972	0.9956	0.9979	0.9951	0.9940
λ (d)	3.45 ± 0.30^{C}	3.59 ± 0.30^{C}	3.57 ± 0.40^{C}	4.85 ± 0.27^{B}	7.88 ± 0.45 ^{A}	8.68 ± 0.49 ^{A}
Pmax (mL CH_4/g-VS)	265.36 ± 3.04	236.53 ± 2.72	259.33 ± 4.44	219.48 ± 2.59	287.07 ± 9.29	250.60 ± 9.73
Rmax (mL CH_4/g-VS·d)	12.56 ± 0.37 ^{A}	11.39 ± 0.35^{B}	11.14 ± 0.41^{B}	10.01 ± 0.27^{CD}	10.46 ± 0.41 ^{BC}	9.05 ± 0.39 ^{D}

* Data in the table are expressed as Means ± SD, and the different letters (A, B, C, …) indicate significant difference at the 5% level ($p < 0.05$).

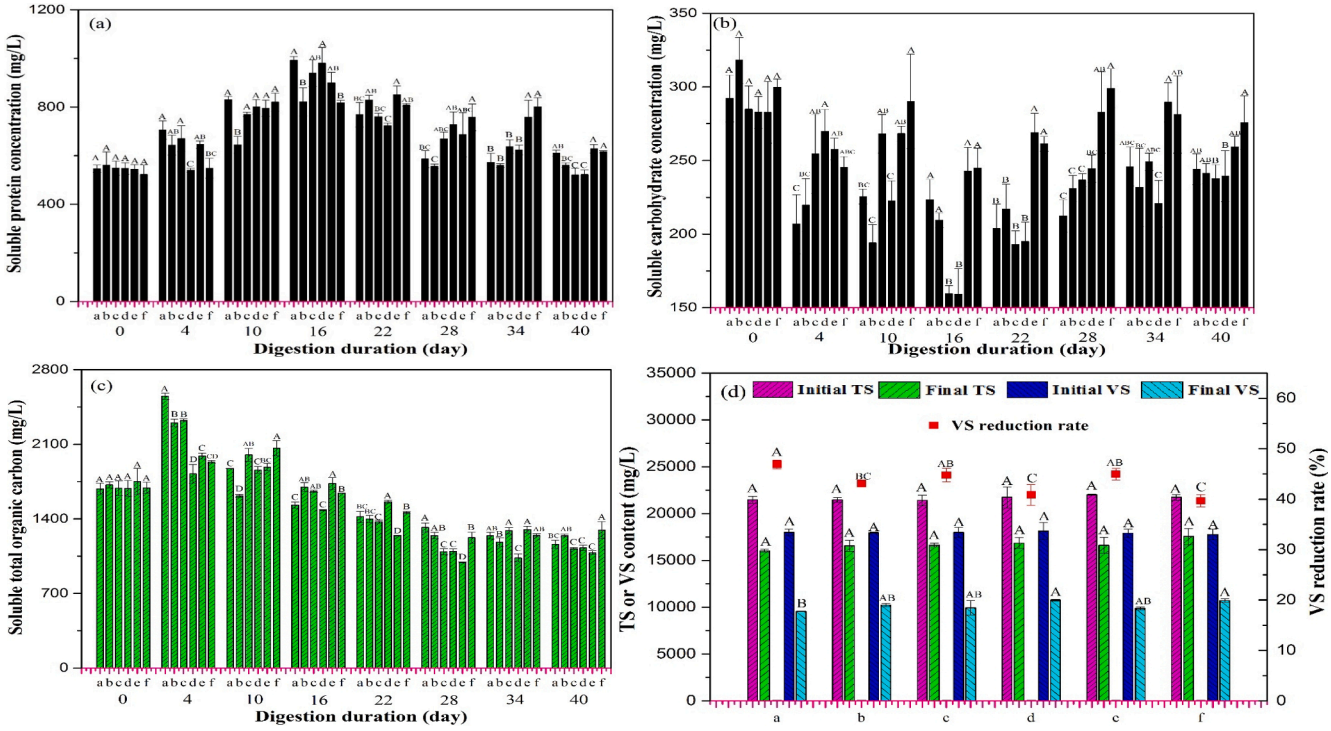


Fig. 5. (a) Soluble protein concentration. (b) Soluble carbohydrate concentration. (c) Soluble total organic carbon. (d) TS or VS and VS reduction rate. The alphabets under X-axis: a-R(6.5 pH)$_{NBW}$, b-R(6.5 pH)$_{Control}$, c-R(6.0 pH)$_{NBW}$, d-R(6.0 pH)$_{Control}$, e-R(5.5 pH)$_{NBW}$, f-R(5.5 pH)$_{Control}$.

the corresponding $R_{Control}$ ($p < 0.05$). In terms of cumulative CH_4 yield (Fig. 4b), R(6.5 pH)$_{NBW}$, R(6.0 pH)$_{NBW}$ and R(5.5 pH)$_{NBW}$ achieved 257 ± 4, 248 ± 4 and 246 ± 6 mL/g-VS, respectively, corresponding to 12.1%, 17.2% and 16.8% increase when compared with R(6.5 pH)$_{Control}$, R(6.0 pH)$_{Control}$ and R(5.5 pH)$_{Control}$, respectively. Similarly, R_{NBW} also showed a significantly higher cumulative CH_4 yield relative to the corresponding $R_{Control}$ ($p < 0.05$).

In order to better analyze the kinetics of the AD process of PM with NBW or DW supplementation under three different degrees of acidic inhibition conditions, the modified Gompertz model was applied and the relevant parameters are shown in Table 2. The results showed that the experimental data well fitted the modified Gompertz model ($R^2 = 0.994$ to 0.998). The difference in the lag phase between $R_{Control}$ and R_{NBW} was not statistically significant ($p > 0.05$). However, the *R*max of R_{NBW} was significantly larger than the corresponding $R_{Control}$ ($p < 0.05$), implying that NBW supplementation increase the maximum CH_4 production rate even under acid inhibition conditions.

3.3. Soluble organics performance

The changes in soluble proteins and carbohydrates in this work are shown in Fig. 5a and Fig. 5b. The soluble proteins in all reactors experienced a trend of increasing and then decreasing. The initial increase may be because hydrolysis promoted the production of soluble proteins, and the consumption rate of soluble proteins was slower than productivity. Regarding the change of soluble proteins on the 4th day, the increments of R_{NBW} were faster than those of $R_{Control}$, but the difference was not statistically significant ($p > 0.05$). With the progress of AD, the production rate of soluble proteins was lower than the consumption rate, and its content gradually decreased. For the soluble carbohydrates on the 4th day, R(6.5 pH)$_{NBW}$ were significantly lower than those of R(6.0 pH)$_{NBW}$, R(6.0 pH)$_{Control}$ and R(5.5 pH)$_{NBW}$ ($p < 0.05$), suggesting that with the deepening of acidic inhibition, the consumption rate of soluble carbohydrates became slower.

Fig. 5c showed the changes of soluble TOC in 40 days' AD with the supplementation of NBW or DW under acidic inhibition conditions. Obviously, the soluble TOC showed a tendency to increase and then decrease. At the beginning of the AD process at day 0, the soluble TOC content of all reactors was basically the same. The difference was not statistically significant ($p > 0.05$). As the AD process progressed, on the 4th day, the soluble TOC content of all reactors increased, which were consistent with the changes in VFAs (Fig. 2). Among them, the soluble TOC content of R(6.5 pH)$_{NBW}$ was significantly higher than that of R(6.0 pH) and R(5.5 pH) ($p < 0.05$). With the progress of AD, the soluble TOC content of all reactors gradually decreased.

The changes in TS and VS content were shown in Fig. 5d. The initial TS and VS contents were 21.64 ± 0.38 and 16.71 ± 0.84 g/L, respectively. The difference in the initial TS and VS contents was not statistically significant (p greater than 0.05). The final VS content of R(6.5 pH)$_{NBW}$ was significantly lower than that of R(6.0 pH)$_{Control}$ and R(5.5 pH)$_{Control}$ ($p < 0.05$). In terms of the VS reduction rate, R(6.5 pH)$_{NBW}$, R(6.0 pH)$_{NBW}$ and R(5.5 pH)$_{NBW}$ were 46.9%, 44.8% and 45.0%,

Table 3
Main characteristics of NBW and DW used in this study.*

Characteristics	Unit	Air-NBW	DW
NBs size	nm	203.00 ± 15.20	ND**
NBs density	10^8 particles/mL	1.10 ± 0.10	ND**
Zeta potential	mV	−16.71 ± 0.60	−1.64 ± 0.13
pH	–	6.71 ± 0.03	6.61 ± 0.03

* Data in the table are expressed as Means ± SD.
** ND (Not Determined).

respectively, corresponding to 43.1%, 40.9% and 39.7% in R(6.5 pH)$_{Control}$, R(6.0 pH)$_{Control}$ and R(5.5 pH)$_{Control}$, respectively. Moreover, the VS reduction rate of R_{NBW} was significantly higher than the corresponding $R_{Control}$ ($p < 0.05$). These phenomena indicated that despite the acidic inhibition, the supplementation of NBW might increase the VS removal rate, which was conducive to the conversion of more organics into CH_4 and other biomass energy.

3.4. Related possible mechanisms and application potential

In this work, multiple parallel tests indicated that under acidic inhibition conditions, the supplementation of NBW would promote CH_4 production (Fig. 4) and digestion stability (Fig. 1 and Fig. 3a) during AD processes. To further clarify the related mechanisms, the characteristics of NBW and DW were also determined. As shown in Table 3, the zeta potential (absolute value) of NBW was significantly higher than DW ($p < 0.05$). As known, zeta potential is an electric potential generated at the solid–liquid interface in response to the relative movement of solid particles and liquid (Cacua et al., 2019). Zeta potential plays an important role in the stability of the bubble (Parmar and Majumder, 2013). In addition, higher zeta potential may lead to higher moving speed, which may also be a reason for the higher mobility of NBW (Takahashi, 2005). Therefore, the higher zeta potential of NBW may cause high water molecule mobility, thereby enhancing the AD process. Moreover, a previous study showed that NBW has higher spin–spin relaxation times (T_2) compared to DW (Wang et al., 2020). The larger the T_2, the more susceptible the hydrogen-bonding network is to interference, thus the higher the mobility of water molecules (Liu et al., 2013). Therefore, the water molecule mobility of R_{NBW} was higher than that of $R_{Control}$.

As reported by Wang et al. (2019), hundreds of millions of NBs could exist stably in NBW for more than 14 days. Michailidi et al. (2020) reported that the NBs produced have long-term stability, up to three months. Moreover, after 35 days, the mobility of NBW was still higher than that of DW (Wang et al., 2020). Therefore, compared with $R_{Control}$, the reactors of R_{NBW} were more likely to have high mobility during the entire AD process rather than only in a shorter time. Therefore, combining the characteristics of NBW and the results of this work, the possible mechanisms of NBW supplementation in AD of PM under acidic inhibition conditions are shown in Fig. 6. At the beginning of the AD process, NBW with higher NBs density, zeta potential (absolute value), T_2 value, and water molecule mobility than DW was added into the reactors. Previous studies have shown that NBW enhances the activity of related hydrolases in the hydrolysis stage (Yang et al., 2019, Fan et al., 2020b). As the fermentation progressed, compared to $R_{Control}$, the generation/consumption of VFAs was enhanced in R_{NBW}, which indicated that the supplementation of NBW facilitated the alleviation of acidic inhibition in the acidification stage. After that, during the methanogenesis stage, NBW supplementation enhanced the CH_4 production in the methanogenesis stage. Therefore, the higher water molecule mobility due to NBW supplementation enhances the entire AD process even under acidic inhibition.

This work demonstrated that under acidic inhibition conditions, NBW still promote CH_4 production. In this work, the increase in CH_4 production in the AD of PM was relatively low, ranging from 12.1% to 17.2%, which was lower than the 21.5% increase achieved by the supplementation of ferrous chloride (Lu et al., 2020) under normal conditions, and also lower than the 36.51% increase achieved by lignite supplementation (Guo et al., 2020a) under normal conditions. However, it is important that the increase in CH_4 production achieved in this work was in acidic inhibition conditions, not in normal conditions. When NBW was used in the AD process of PM under acidic inhibition conditions, although the supplementation of NBW did not increase the CH_4 production effectively, it promoted the digestion stability and played an important role in alleviating acidic inhibition. Moreover, as a pollution-free, chemical-free, and low-cost technology, the manufacture of NBW did not require complex devices or consume too much energy. With the continuous development of NBW technology, NBW may not only significantly alleviate acidic inhibition and improve AD system stability, but also greatly increase CH_4 production in the future. Moreover, the application of NBW technology could not only effectively treat acidic substrates, but also avoid the use of chemicals, which was conducive to reducing the use of chemicals. Therefore, in the AD process under acidic inhibition, NBW technology may have huge potential in the future and is worthy of further research.

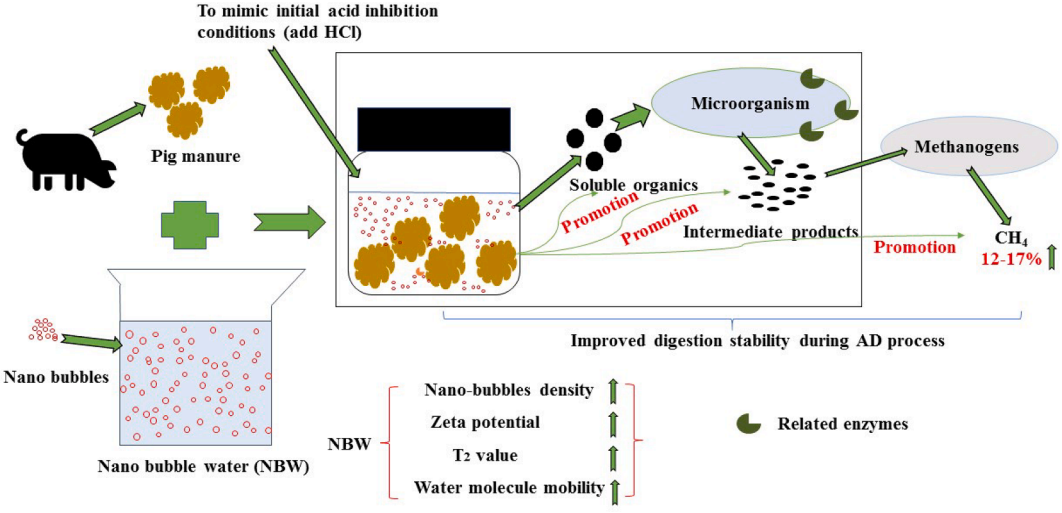


Fig. 6. The possible mechanisms of adding NBW in AD of PM under acid inhibition conditions.

4. Conclusions

This study investigated the ability of NBW in promoting AD recovery from acidic inhibition. Results demonstrated that stability of the AD process suffered acidic inhibition was maintained with the NBW supplementation, due to the elevation of TA and PA, as well as the declining ratio of VFA/TA. Besides, generation/consumption of VFAs was accelerated with NBW supplementation. Moreover, 12.1–17.2% higher cumulative CH_4 yield were achieved in NBW groups when compared to the corresponding controls. The above results showed an enhanced recoverability of NBW-based AD system from acidic inhibition, suggested the promising application of NBW for the alleviation of extreme conditions.

CRediT authorship contribution statement

Yujie Fan: Investigation, Methodology, Writing - original draft, Data curation. **Xiaojing Yang:** Formal analysis, Methodology. **Zhongfang Lei:** Conceptualization, Supervision. **Yasuhisa Adachi:** Resources. **Motoyoshi Kobayashi:** Resources. **Zhenya Zhang:** Conceptualization, Supervision. **Kazuya Shimizu:** Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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